25PY101: Engineering Physics

Dr. Sreekar Guddeti

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References

post M2T1

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1 Topics

Topics	Avadhanulu	Neamen
Fibre Optics		
	47.1 Introduction	
	47.2 Optical Fibre	
	47.3 Total Internal Reflection	
	47.4 Propagation of Light and Optical Fibre	
	47.5 Fractional Refractive Index Change	
	47.6 Numerical Aperture	
	47.20 Fibre Optic Communication System	
Lasers		
	44.1 Introduction	14.6 Laser Diodes
	44.2 Interaction of light with matter and three quantum processes	
	44.4 Light Amplification	
	44.5 Meeting the three requirements	
	44.6 Components of laser	
	44.7 Lasing action	
	44.11.5 Semiconductor diode laser	
	44.13 Applications	

2 Test your understanding

The questions marked ■ are important conceptual questions. The questions marked ■ involve sketching and are also important.

Fibre Optics

1. What is an optical fibre? What is the principle involved in its working?

2. What are optical fibres? Give its applications.
3. Explain core, cladding and sheath in optical fibre by drawing a neat diagram.
4. Explain the phenomenon of total internal reflection. How is it used in fibre optic communication?
5. What is meant by critical propagation angle of an optical fibre? Obtain an expression for critical propagation angle.
6. What is meant by critical angle of an optical fibre? Obtain an expression for critical angle.
7. What is meant by acceptance angle of an optical fibre? Obtain an expression for acceptance angle.
8. What is meant by fractional refractive index of an optical fibre? Obtain an expression for fractional refractive index.
9. What is meant by numerical aperture of an optical fibre? Obtain an expression for numerical aperture.

Lasers

1. How is laser light different from ordinary light?
2. Explain with neat diagram, the processes of absorption of light, spontaneous emission, and stimulated emission of light. What are the necessary conditions for their occurrence?
3. Why does spontaneous emission dominate over stimulated emission at normal temperature?
4. Explain the terms: Spontaneous and stimulated emission. Which of the two emission processes is maximized for laser operation?
5. Explain in laser: (i) Why the active medium should have broad absorption band? (ii) What is the role of metastable state?
6. What are the three requirements for lasing action?
7. Using neat diagram, describe the three components of a laser.
8. Using neat diagram, describe the Fabry-Perot optical resonator.
9. Using neat diagrams, describe the lasing action?
10. Describe the operation of a laser diode.
11. In a semiconductor diode laser, what is the active medium?
12. In a semiconductor diode laser, what is the pumping agent?
13. In a semiconductor diode laser, what is the optical resonator?
14. Using neat diagram, describe the homojunction diode laser and its components.
15. Using neat diagram, describe the heterojunction diode laser and its components.
16. Using neat diagram, explain optical fibre communication system and the role of semiconductor diode laser.

3 Technical terms

Fibre Optics

- optical fibre
- total internal reflection
- core, cladding, sheath
- critical angle of an optical fibre
- critical propagation angle of an optical fibre
- fractional refractive index
- acceptance angle of an optical fibre
- Long haul communication

Lasers

- LASER
- Stimulated absorption
- Spontaneous emission
- Stimulated emission
- Population inversion
- metastable state
- optical resonator
- Fabry-Perot optical resonator
- Optical pumping
- active medium
- Optical resonant cavity or optical resonator
- Lasing action
- Long haul communication